

ARUN KUMAR REDDY N

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SUMMARY

7+ years optimizing **computer vision** and **firmware** across DSPs, GPUs, and hardware accelerators, with a focus on **SIMD/GPU parallelization**

EXPERIENCE

Qualcomm

Optimization Engineer

Feb 2025 - Present

- Accelerated XR algorithms on Hexagon DSP via **HVX SIMD** intrinsics and **Halide** framework
- Built **C++26 std::simd** abstractions for low-level intrinsics, improving team adoption and hardware utilization
- Resolved hardware bugs to preserve SIMD execution paths, preventing inefficient scalar fallbacks

Firmware Engineer

Sept 2022 - Jan 2025

- Developed EVA **hardware accelerator firmware** from FPGA verification through Mobile, Auto and IoT commercialization
- Re-Architected firmware with dynamic HW scheduling and DMA utilities to maximize hardware utilization
- Led competitive analysis by bringing up **multi-threaded** firmware on Hexagon DSP

Valeo

Optimization Engineer

July 2019 - Sept 2022

- Ported ADAS algorithms to heterogeneous architectures (Renesas DSP, Adreno GPU **OpenCL** & **ARM NEON**)
- Built CMake cross-compilation toolchains to evaluate CV prototypes on Nvidia & Qualcomm SoCs
- Automated processor-in-loop framework for streamlined on-target issue reproduction & debugging

SKILLS

C, C++ | SIMD (HVX, NEON) | DSP (Renesas CVe, Qualcomm Hexagon) | GPGPU (OpenCL, CUDA) | Firmware | Multi-Threading

EDUCATION

Ramaiah Institute Of Technology

B.E. (Electrical & Electronics Engineering)

Aug 2015 - June 2019